AI

TUTORIAL : 01

NAME: MONIKA KAPHLE

STUDENT ID: 2408878

GROUP: 18

AI

Tutorial :01

1. 4.)1. Suppose that :

$A = \{x : x \in N$ and x is even$\}$,

$B = \{x : x \in N$ and x is prime$\}$, and

$C = \{x : x \in N$ and x is a multiple of $5\}$.

a. $A \cap B$

Ans: Numbers which are both even and prime.

b. $B \cap C$

Ans: Numbers which are both prime and multiple of 5.

c. $A \cup B$

Ans: Numbers which are even OR prime. (Union)

d. $A \cap (B \cup C)$

Ans: $B \cup C$ - Prime or multiple of 5 then common with even numbers.

2.) Prove:

i. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Solution; Here.

First Inclusion : $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Taking the left side :

$x \in A \cup (B \cap C)$ which means,

$x \in A$ or $x \in B \cap C$

Case 1 . $x \in A$

x is in $A \cup B$ and $A \cup C$ (both sets).

$\therefore x \in (A \cup B) \cap (A \cup C)$.

case 2: $x \in B \cap C$ ($x \in B$ and $x \in C$)
x is in $A \cup B$ and x is in $A \cup C$ (both def)
So, $x \in (A \cup B) \cap (A \cup C)$
$\therefore A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$

- Second Inclusion: $(A \cup B) \cap (A \cup C) = A \cup (B \cap C)$

Taking the left side:
$x \in (A \cup B) \cap (A \cup C)$ which means $x \in (A \cup B)$ and $x \in A \cup C$.

case 1: $x \in A \cup B / x \in A$
x is in $A \cup (B \cap C)$

case 2: $x \notin A / x \in B$
$x \in A \cup B$ and $x \notin A$ it must be $x \in B$.

Similarly, since $x \in A \cup C$ and $x \notin A$ it must be $x \in C$.
$\because x \in B \cap C$, so, $x \in A \cup (B \cap C)$.
Since, both case imply that $x \in (A \cup (B \cap C))$ which shown that $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$.

Conclusion:
From both first and second inclusion we can conclude that:
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Here;

- First Inclusion: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Taking the left side:
$x \in A \cap (B \cup C)$ which means
$x \in A$ and $x \in B \cap C$

case 1: $x \in B$
x is in $A \cap B$

Case 2: $x \in C$

x is in $A \cap C$

$\therefore x \in (A \cap B) \cup (A \cap C)$

2.1 Matrix, Operations and system of linear Eq^n.

1. a. Write down 3x3 matrix with one on diagonal and zero.

Required matrix: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

b. Write down 3x4 matrix with one in diagonal and zero.

Required matrix: $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$.

c. Write down 4x3 matrix with one in diagonal and zero.

Required matrix: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

2. let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ Compute AD and DA.

Here;

$AD = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

$$= \begin{bmatrix} 1\times2+1\times0+1\times0 & 1\times0+1\times3+1\times4 & 1\times0+1\times0+1\times4 \\ 1\times2+2\times0+3\times0 & 1\times0+2\times3+3\times0 & 1\times0+2\times0+3\times4 \\ 1\times2+3\times0+4\times0 & 1\times0+3\times3+4\times0 & 1\times0+3\times0+4\times4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 & 4 \\ 2 & 6 & 12 \\ 2 & 9 & 16 \end{bmatrix}$$

$$DA = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2\times1+0\times1+0\times1 & 2\times1+0\times2+0\times3 & 2\times1+0\times3+0\times4 \\ 0\times1+3\times1+0\times1 & 0\times1+3\times2+0\times3 & 0\times1+3\times3+0\times4 \\ 0\times1+0\times1+4\times1 & 0\times1+0\times2+4\times3 & 0\times1+0\times3+4\times4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 2 \\ 3 & 6 & 9 \\ 4 & 12 & 16 \end{bmatrix}$$

From observation.

$AD \neq DA$, so matrix multiplication is not commutative as D. is a diagonal matrix.

. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$

Verify that $AB = AC$

Here;

$$AB = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1\times2+2\times1 & 1\times1+2\times3 \\ 2\times2+4\times1 & 2\times1+4\times3 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

$$AC = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1\times4+2\times0 & 1\times3+2\times2 \\ 2\times4+4\times0 & 2\times3+4\times2 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

$$\therefore AB = AC = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

4. Prove that: $(AB)^T = B^T A^T$

Solution; Here;

let,

A be an $m\times n$ matrix

B be an $n\times p$ matrix.

Then;

AB is an $m\times p$ matrix

Transpose:

$$(AB)^T_{ij} = (AB)_{ji}$$

But;

$$(AB)_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$$

So,

$$(AB)^T_{ij} = \sum_{k=1}^{n} a_{jk} b_{ki} \rightarrow eq^n \text{ (i)}$$

now, computing $B^T A^T$

For A:

$$(A^T)_{kj} = a_{jk}$$

For B:

$$(B^T)_{ik} = b_{ki}$$

Multiplying $B^T A^T$

Then ij element of B^TA^T is

$$(B^TA^T)_{ij} = \sum_{k=1}^{n} (B^T)_{ik}(A^T)_{kj}$$

Substituting transpose definitions:

$$(B^T)_{ik} = b_{ki}$$

$$(A^T)_{kj} = a_{jk}$$

So, $(B^TA^T)_{ij} = \sum_{k=1}^{n} b_{ki}\, a_{jk} \rightarrow$ eq^n (ii)

Computing both expression (i) and (ii)

both eq^n are commutative

$$a_{jk}b_{ki} = b_{ki}\, a_{jk}$$

Thus,

$$(AB)^T_{ij} = (B^TA^T)_{ij}$$

Hence proved.

5. $A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$ Show that is a symmetric square matrix. Observe sum of A^TA is sum of squares of elements of A.

Here;

Transpose of $A(A^T) = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

$$A^TA = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

$$= \begin{bmatrix} a\times a + b\times b + c\times c & a\times d + b\times e + c\times f \\ d\times a + e\times b + f\times c & d\times d + e\times e + f\times f \end{bmatrix}$$

$$= \begin{bmatrix} a^2+b^2+c^2 & ad+be+cf \\ ad+be+cf & d^2+e^2+f^2 \end{bmatrix}$$

This matrix is symmetric.

6. And inverse of matrix $A = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$ $B = \begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix}$

Here; Given;

$$A^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{\text{def}(A)} \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

now;

$$A = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$$

$$|A| = 5 \times 5 - 6 \times 4 = 1.$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$$

$$|B| = 6 \times 3 - 4 \times 3 = 6$$

$$B^{-1} = \frac{1}{6} \begin{bmatrix} 3 & -4 \\ -3 & 6 \end{bmatrix} = \begin{bmatrix} 1/2 & -2/3 \\ -1/2 & 1 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix} \text{ and } B^{-1} = \begin{bmatrix} 1/2 & -2/3 \\ -1/2 & 1 \end{bmatrix}$$

7. Given matrices A and B. verify they are inverse:

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$$

Here;

Two matrices A and B are inverse of if

$AB = I$ and $BA = I$ where $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

now;

$$AB = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 4 \times 1 + 1 \times (-3) & 4 \times (-1) + 1 \times 4 \\ 3 \times 1 + 1 \times (-3) & 3 \times (-1) + 1 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$BA = \begin{pmatrix} 1 & -1 \\ -3 & 4 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 1\times4+(-1)\times3 & -1\times1+(-1)\times1 \\ (-1)\times4+4\times3 & (-3)\times1+4\times1 \end{pmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

now, $AB = BA = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

8. Verify that given matrices are inverse of each other

$$\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix}$$

Solution, Here;

Let $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix}$

To be inverse of each other,

$AB = BA = I$ where $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

now,

$$AB = \begin{bmatrix} 1\times3+(-1)\times2+0\times3 & 1\times(-4)+(-1)\times(-4)+0\times(-5) & 1\times1+(-1)\times1+0\times1 \\ 1\times3+0\times2+(-1)\times3 & 1\times(-4)+0\times(-4)+(-1)\times(-5) & 1\times1+0\times1+(-1)\times1 \\ 2\times3+3\times2+(-4)\times3 & 2\times(-4)+3\times(-4)+(-4)\times(-5) & 2\times1+3\times1+(-4)\times1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Since,

$$AB = BA = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ proved.}$$

3.4.1 System with two variable.

a. $\begin{cases} x+2y = 4 \\ 3x-5y = 1 \end{cases}$

Here, Given;

$x+2y = 4 \quad \ldots \text{ eq}^n \text{ (i)}$

$3x-5y = 1 \quad \ldots \text{ eq}^n \text{ (ii)}$

now,

Converting into Matrix form;

$A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}$ $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $B = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ which means

i's $AX = B$.

Determinant of $A(|A|) = 1\times(-5) - 3\times2 = -5-6 = -11 \neq 0$

We know;

$$A^{-1} = \frac{1}{|A|} \times \text{Adj of } A.$$

$$= \frac{1}{-11} \times \begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix}$$

To find $X = A^{-1}B$

$$= \frac{1}{-11} \begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$= \frac{1}{-11}\begin{bmatrix} -5\times4 & -2\times1 \\ -3\times4 & +1\times1 \end{bmatrix}$$

$$= \frac{1}{-11}\begin{bmatrix} -22 \\ -11 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$\therefore x = 2$ and $y = 1$.

b. $\begin{cases} 5x + y = 13 \\ 3x + 2y = 5. \end{cases}$

Here, Given;

$5x + y = 13$ ..... eq^n (i)

$3x + 2y = 5$ ..... eq^n (ii)

now;

converting into Matrix form,

$A = \begin{bmatrix} 5 & 1 \\ 3 & 2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ $B = \begin{bmatrix} 13 \\ 5 \end{bmatrix}$ which is $AX = B$.

Determinant of A $(|A|) = 5\times2 - 1\times3 = 7$.

we know;

$$A^{-1} = \frac{1}{|A|} \times \text{Adj of } A$$

$$= \frac{1}{7} \times \begin{bmatrix} 2 & -1 \\ -3 & 5 \end{bmatrix}$$

To find $X = A^{-1}B$

$$= \frac{1}{7}\begin{bmatrix} 2 & -1 \\ -3 & 5 \end{bmatrix}\begin{bmatrix} 13 \\ 5 \end{bmatrix}$$

$$= \frac{1}{7}\begin{bmatrix} 2\times13 & -1\times5 \\ -3\times13 & +5\times5 \end{bmatrix}$$

$$= \frac{1}{7}\begin{bmatrix} 21 \\ -14 \end{bmatrix}$$ $\therefore x = 3$ and $y = -2$.

e. $\begin{cases} 4x+3y=11 \\ x-3y=-1. \end{cases}$

Here, Given;

$4x+3y=11 \rightarrow eq^n$ (i)

$x-3y=-1 \rightarrow eq^n$ (ii)

now,

converting into Matrix form

$A = \begin{bmatrix} 4 & 3 \\ 1 & -3 \end{bmatrix}$ $X = \begin{bmatrix} x \\ y \end{bmatrix}$ $B = \begin{bmatrix} 11 \\ -1 \end{bmatrix}$ which is $AX = B$.

Determinant of $A(|A|) = 4 \times (-3) - 3 \times 1 = -15 \neq 0$.

we know,

$A^{-1} = \frac{1}{|A|} \times$ Adj of A.

$= \frac{1}{-15} \times \begin{bmatrix} -3 & -3 \\ -1 & 4 \end{bmatrix}$

To find $X = A^{-1}B$

$= \frac{1}{-15} \begin{bmatrix} -3 & -3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 11 \\ -1 \end{bmatrix}$

$= \frac{1}{-15} \begin{bmatrix} (-3) \times 11 + 3 \times 1 \\ (-1) \times 11 - 4 \times 1 \end{bmatrix}$

$= \frac{1}{-15} \begin{bmatrix} -30 \\ -15 \end{bmatrix}$

$= \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$\therefore x = 2$ and $y = 1$.

3.4(2) System with three variables.

a. $\begin{cases} x+y-z = 2 \\ x+z = 7 \\ 2x+y+z = 13 \end{cases}$

Here,

converting into augmented Matrix;

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 7 \\ 13 \end{bmatrix} = [A|B]$$

Performing elementary Row operation until we reached Row echelon form:

$\rightarrow R_2 \rightarrow R_2 - R_1$

$= [1-1, 0-1, 1-(-1) | 7-2] = [0, -1, 2 | 5]$

$\rightarrow R_3 \rightarrow R_3 - 2R_1$

$= [2-2, 1-2, 1-(-2) | 13-4] = [0, -1, 3, 9]$

new matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & 5 \\ 0 & -1 & 3 & 9 \end{array}\right]$$

$R_3 \rightarrow R_3 - R_2$

$= (0-0, -1-(-1), 3-2 | 9-5] = [0, 0, 1 | 4]$

new matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & 5 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

now, Row Echelon form :

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} \quad \ldots eq^n \ (i)$$

writing eq^n (i) as linear combination

$$x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix}$$

which in eq^n is $\left\{ \begin{array}{l} x + y - z = 2 \\ -y + 2z = 5 \\ z = 4 \end{array} \right\} \ldots eq^n$ (ii)

$\therefore z = 4$

from eq^n (ii) $-y + 2z = 5$

or, $-y + 2 \times 4 = 5$

or, $-y = 5 - 8$

$\therefore y = 3$

from eq^n (ii) $x + y - z = 2$

or, $x + 3 - 4 = 2$

$\therefore x = 3$

$\therefore x = 3, y = 3$ and $z = 4$.

b.) $\left\{ \begin{array}{l} x + y - z = 2 \\ 3x + y = 7 \\ x + y + 2z = 3 \end{array} \right.$

Solution: Here;

writing in Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 3 & 1 & 0 & 7 \\ 1 & 1 & 2 & 3 \end{array}\right] = [A|B]$$

Performing elementary Row Operations.

$R_2 \to R_2 - 3R_1$

$[3-3,\ 1-3,\ 0-(-3) \mid 7-6] = [0, -2, 3 \mid 1]$

$R_3 \to R_3 - R_1$

$[1-1,\ 1-1,\ 2-(-1) \mid 3-2] = [0, 0, 3 \mid 1]$

3.4.14.

c. $\begin{cases} 3x+2y = -2 \\ x+4y = 6 \end{cases}$

Solution;

$|A| = \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} = 3\times4 - 1\times2 = 12-2 = 10$

$A^{-1} = \frac{1}{10}\begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$

$x = A^{-1}B = \frac{1}{10}\begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} -2 \\ 6 \end{bmatrix} = \begin{bmatrix} 4\times(-2) - 2\times6 \\ -1\times(-2) + 3\times6 \end{bmatrix} = \begin{bmatrix} -8-12 \\ 2+18 \end{bmatrix}$

$= \frac{1}{10}\begin{bmatrix} -20 \\ 20 \end{bmatrix}$

$= \begin{bmatrix} -2 \\ 2 \end{bmatrix}$

$\therefore x = -2,\ y = 2.$

d. $2x+3y = 6$

$x - y = 1/2$

Solution:

Here;

$|A| = \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} = 2\times(-1) - 1\times3 = -2-3 = -5.$

$A^{-1} = \frac{1}{-5}\begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}$

$X = A^{-1}B = \frac{1}{-5}\begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 6 \\ 1/2 \end{bmatrix}$

$= \frac{-1}{5}\begin{bmatrix} -1\times6 - 3\times1/2 \\ -1\times6 + 2/2 \end{bmatrix}$

$= \begin{bmatrix} 5/7 \\ 1/7 \end{bmatrix}$

f. $2x+3y=8$

$4x+6y=20$

Soln. $|A| = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 12-12 = 0$ $\therefore$ No solution.

2 b. $\begin{cases} -x-2y+z=1 \\ 2x+3y=2 \\ y-2z=0 \end{cases}$

Solution;

$y=2z$

$2x+3y=2$

or, $2x+3\times2z=2$

or, $2x+6z=2$

or, $2x-2+6z=0$ Divided by 2, $x+3z=1$.

Now;

(i) $-x-2y+z=-1$

or, $-2y+z=x$

or, $-2\times2z+z=x$

or, $-4z+z=x$ $\therefore$ $x+3z=-1$

From, $x+3z=1$

$x+3z=-1$

$\therefore z=1.$

$\therefore y = 2\times1 = 2$

or, $x - 3\times1 = 1$

$\therefore x = -1$.

d. $x+4y-z=4$

$2x+5y+8z=15$

$x+3y-3z=1$.

Solution;

Subtracting eqⁿ (iii) from (i)

$(x+4y-z)-(x+3y-3z) = 4-1$

$\therefore y+2z=3$

Subtracti eqⁿ (iii) from (ii)

$(2x+5y+8z)-(x+3y-3z) = 15-1$

$\therefore x+2y+11z=14$

Now, $y = 3-2z$

$x+2(3-2z)+11z=14$

or, $x+6-4z+11z=14$

$\therefore x+7z=8$

Now, $x+3(3-2z)-3z=1$

or, $x+9-6z-3z=1$

or, $x-9z=-8$

From, $x+7z=8$

$x-9z=-8$

or, $16z=16$ $\therefore z=1$.

$\therefore y = 3-2\times1 = 1$

$\therefore x+7\times1 = 8$ $= -1$. Ans.

e. $5x+3y+9z=-1$
$-2x+3y-z=-2$
$-x-4y+5z=1$
Solution:
Sub equation (ii) from (i)
$(5x+3y+9z)-(-2x+3y-z)=-1+2$
$\therefore 7x+10z=1$
Now, Add eq^n (ii) and (iii)
$-2x+3y-z-x-4y+5z=-2+1$
$\therefore -3x-y+4z=-1$
From, solving $z=0$
So,
eq^n (i) $5x+3y=-1$
(ii) $-2x+3y=-2$
Subtracting, $7x=1$
$x=1/7$.

f. $6x-5y+2z=3$
$2x+y-4z=5$
$3x-3y+z=-1$
Solution;
Multiplying equation (ii) by 3
$6x+3y-12z=15$
Now, subtracting equation (i)
$8y-14z=12$
Now, multiply (iii) by 2
$6x-6y+2z=-2$
Now, subtracting equation (i) $y=5$.
Using equation (ii), $2x+5-4z=5$

or, $2x-4z=0$
$\therefore x = 2z$
or, $3(2z)-3\times5+z=-1$
or, $6z-15+z=-1$
or, $7z=14$
$\therefore z=2$
$\therefore x=4$
$\therefore y=5.$

4.1.1 i) $f(x)=x+\sqrt{x}$

$$f(x)=\frac{d}{dx}(x+\sqrt{x})$$
$$=\frac{dx}{dx}+\frac{d\sqrt{x}}{dx}$$
$$=1+\frac{d}{dx}(x^{1/2})$$
$$=1+\frac{1}{2}x^{1/2-1}$$
$$=1+\frac{1}{2}x^{-1/2}$$
$$=1+\frac{1}{2\sqrt{x}}$$ Ans.

2. $f(x)=2+\frac{1}{x}+\frac{1}{x^2}$

or, $f(x)=2+x^{-1}+x^{-2}$

$$f'(x)=\frac{d}{dx}(2)+\frac{d}{dx}(x^{-1})+\frac{d}{dx}(x^{-2})$$
$$=0-x^{-2}-2x^{-3}$$

$= \frac{-1}{x^2} - \frac{2}{x^3}$ Ans.

3. $f(x) = \sqrt{x} \cos x$

let $u = \sqrt{x} = x^{1/2}$ $v = \cos x$

Now,

$\frac{dv}{dx} = \frac{d}{dx}(x^{1/2}) = \frac{1}{2\sqrt{x}}$

$\frac{dv}{dx} = \frac{d}{dx}(\cos x) = \sin x$

$\therefore f'(x) = \frac{1}{2\sqrt{x}} \cos - \sqrt{x} \sin x$

4. $f(x) = x^3 e^x$

let $u = x^3$, $v = e^x$

$\frac{du}{dx} = \frac{dx^3}{dx} = 3x^2$

$\therefore f'(x) = 3x^2 e^x - x^3 e^x = e^x(3x^2 - x^3)$ Ans.

5. $f(x) = \frac{x^3}{e^x}$

or, $f(x) = x^3 e^{-x}$

let $u = x^3$, $v = e^{-x}$

$\frac{du}{dx} = 3x^2$, $\frac{dv}{dx} = -e^{-x}$

$\therefore f'(x) = 3x^2 e^{-x} - x^3 e^{-x}$ Ans.

6. $f(x) = (3x^2+2x+1)^5$

let $u = 3x^2+2x+1$

so, $f(x) = u^5$

$\frac{du}{dx} = \frac{d}{dx}(u^5) = 5u^4 \cdot u'$

$u' = 6x+2$.

$\therefore f'(x) = 5(3x^2+2x+1)^4\,(6x+2)$.

7. $f(x) = (\sin x)^4$

let $u = \sin x$

Then, $f(x) = u^4$

or, $\frac{du}{dx} = \frac{d(u^4)}{dx} = 4u^3 \cdot u'$

or, $u' = \cos x$

$\therefore f'(x) = 4\sin^3(x)\cos x$.

8. $f(x) = e^{(3x^2+2x)}$

let $u = 3x^2+2x$

Then, $f(x) = e^u$

so, $\frac{du}{dx} = \frac{d}{dx}(e^u) = e^u \cdot u'$

$u' = 6x+2$

$\therefore f'(x) = e^{3x^2+2}\,(6x+2)$

9. $f(x) = \ln(5x^2+3x)$

let $u = 5x^2+3x$

$f(x) = \ln u$

so, $\frac{du}{dx} = \frac{d(\ln u)}{dx} = \frac{u'}{u}$

$u' = 10x + 3$

$\therefore f'(x) = \dfrac{10x+3}{5x^2+3x}.$

10. $f(x) = \sqrt{x^3+4x+1}$

$f(x) = (x^3+4x+1)^{1/2}$

let;

$u = x^3+4x+1$

Then, $f(x) = u^{1/2}$

$f'(x) = \dfrac{1}{2} u^{-1/2} u'$

$u' = 3x^2+4$

$f'(x) = \dfrac{1}{2}(x^3+4x+1)^{-1/2}(3x^2+4)$

$\therefore f'(x) = \dfrac{3x^2+4}{2\sqrt{x^3+4x+1}}$

Problem 2.

Given; $g(x) = x^2 - 3x + 2$

(1) Derivative of $f(x) \cdot g(x)$ at $x = 1$

$g(1) = 1^2 - 3 \times 1 + 2 = 0$

$g'(x) = 2x - 3$

chain rule, $(fg)'(1) = f'(1)\,g(1) + f(1) * g'(1)$

$= 0 \times 0 + 5 \times (-1)$

$= -5.$

As. $f(0) = 3$ and $f'(0) = -1$
$g(0) = 0^2 - 3\times0 + 2 = 2$
$g'(0) = 2\times0 - 3 = -3$

(2.) Derivative of $\frac{f(x)}{g(x)}$ at $x = 0$.

$$\left(\frac{f}{g}\right) = \frac{f'g - fg'}{g^2}$$

$$\left(\frac{f}{g}\right)'(0) = \frac{(-1)\times2 - 3\times(-3)}{2^2} = \frac{-2+9}{4} = \frac{7}{4}.$$

(3.) Derivative of $f(g(x))$ at $x = 0$
$[f(g(x))]' = f'(g(x)).g'(x)$

$\because g(0) = 2$
As. $f'(2) = 3$ and $g'(x) = 2x - 3$
$g'(0) = -3$
Substituting; $[f(g(x))]'_{x=0} = f'(2).(-3)$
$= 3(-3)$
$= -9.$

4.2.1.i) $g(x) = x^2 - 4x + 5$ at Interval $[1, 4]$
$g'(x) = 2x - 4$
$g'(1) = 2\times1 - 4 = 2 - 4 = -2$ $\quad \because g'(4) = 0$
$g(1) = 1 - 4 + 5 = 2$
$g(4) = 16 - 16 + 5 = 5$
$\because$ Min at $g(1)$ and max at $g(4)$

2.) $h(x) = \frac{x^2}{2} - 2x$ at interval $[-2, 3]$

$h'(x) = x - 2$

$\because x = 2$ so, $x = 2$ is critical point.

$h(-2) = \frac{4}{2} + 4 = \frac{4+8}{2} = \frac{12}{2} = 6.$

$h(3) = \frac{9}{2} - 6 = \frac{9-12}{2} = -\frac{3}{2}$

$h(2) = \frac{4}{2} - 4 = 4 - \frac{8}{2} = \frac{-4}{2} = -2.$

$\therefore$ Min at $h(2) = -2$ and max. at $h(-2)$

3. $p(x) = 3x^3 - 9x^2 + 7$ at interval $[-1, 2]$

$p'(x) = 9x^2 - 18x$

$p(-1) = -3 - 9 + 7 = -5.$

$p(2) = 3 \times 8 - 9 \times 4 + 7 = 24 - 36 + 7 = -5$

$p(0) = 7.$

$\therefore$ Min. at $p(-1)$ or $p(2)$ and max. at $p(0)$.

4. $f(x) = x^2 - 4x + 6$ at interval $[0, 3]$

$f'(x) = 2x - 4$

$f(0) = 0 - 0 + 6 = 6$

$f(3) = 9 - 12 + 6 = 9 - 6 = 3.$

Min at $f(3)$ and max at $f(0)$.

$$\frac{\partial m}{\partial y} = \frac{2y}{x^2+y^2+1}$$

$$\nabla m = \left(\frac{2x}{x^2+y^2+1}, \frac{2y}{x^2+y^2+1}\right).$$

a) at $(0,0) = \left(\frac{2\times 0}{0+0+1}, \frac{2\times 0}{0+0+1}\right) = (0,0)$

b) at $(1,-1) = \left(\frac{2\times 1}{1^2+(-1)^2+1}, \frac{2\times(-1)}{1+1+1}\right) = \left(\frac{2}{3}, \frac{-2}{3}\right)$

c) at $(-2,3) = \left(\frac{2\times(-2)}{4+9+1}, \frac{2\times 3}{4+9+1}\right) = \left(\frac{-2}{7}, \frac{3}{7}\right)$

2. Maxima and minima of multivariate function.

1. $m(x,y) = -x^2-y^2+4x+6y$

$\frac{\partial m}{\partial x} = -2x+y = 0 \Rightarrow x=2$

$\frac{\partial m}{\partial y} = -2y+6=0 \Rightarrow y=3$

$m(2,3) = -4-9+8+18 = 13$

$m(0,0)=0$, $m(4,0)=0$, $m(0,5)=5$, $m(4,5)=5$

$m(2,0)=4$, $m(2,5)=5$, $m(0,3)=9$, $m(4,3)=9$

$\therefore$ max at $(2,3)$ and min. at $(0,0)$ and $(4,0)$.

2. $f(x,y) = xy+x^2-4y+6$

$\frac{\partial f}{\partial x} = y+2x=0$

$\frac{\partial g}{\partial y} = x - y = 0 \Rightarrow x = y$ (outside interval)

$g(-2,3) = -6+4-12+6 = -8$

$g(2,1) = 2+4-4+6 = 8$

5. $\cancel{p(x,y) = x^3 - 3xy + 4 \text{ interval}}\ x : [0,2]$ and $y \setminus [0,3]$

$g(2,3) = 6+4-12+6 = 4.$

On edge $x = 2$: $g = -2y+4-4y+16 = 10-6y$ ~~mat~~ max at $y = 1 : g = 4$

min at $y = 3 : g = 8$

$g(-2,4) = -8+64-16+6 = 46.$

3. $p(x,y) = x^3 - 3xy + 4$ interval $x : [0,2]$ and $y = [0,3]$

$\frac{\partial p}{\partial x} = 3x^2 - 3y = 0 \Rightarrow y = x^2$

$\frac{\partial p}{\partial y} = -3x = 0 \Rightarrow x = 0$ then, $y = 0$

$p(0,0) = 4$

$p(2,0) = 8+4 = 12$

$p(0,3) = 4$

$p(2,3) = 8-18+4 = -6$

$\therefore$ Max. at $p(2,0)$ and min at $p(2,3)$.

4. $h(x,y) = 2x^2 + 3y^2 - 12x - 18y + 20$

$\frac{\partial h}{\partial x} = 4x - 12 = 0 \Rightarrow x = 3.$

$\frac{\partial h}{\partial y} = 6y - 18 = 0 \Rightarrow y = 3.$

$h(3,3) = 18+27-36-54+20 = -25$

$h(5,0) = 2\times25+0-12\times5-0+20 = 10$

$h(1,0) = 2+0-12-0+20 = 10$

$h(1,4) = 2\times1+3\times16-12\times1-18\times4+20 = -14$

$h(5,4) = 2\times25+3\times16-12\times5-18\times4+20 = -14$

$\therefore$ Max at $(1,0)$ and $(5,0)$, min at $(3,3)$.

5. $f(x,y) = x^2+y^2-4x-6y+13$

$\therefore \frac{\partial f}{\partial x} = 2x-4 = 0 \Rightarrow x = 2$

$\therefore \frac{\partial f}{\partial y} = 2y-6 = 0 \Rightarrow y = 3$

$f(2,3) = 4+9-8-18+13 = 0$

$f(1,1) = 1+1-4-6+13 = 5$

$f(3,1) = 9+1-12-6+13 = 5$

$f(1,4) = 1+16-4-24+13 = 2$

$f(3,4) = 9+16-12-24+13 = 2$

$\therefore$ Max at $f(1,1)$ and $f(3,1)$

$\therefore$ Min at $(2,3)$.